

1. Dataset Loading and Preview

localhost:8888/notebooks/Documents/Employee%20Burnout%20Analysis%20Project/Project_analysis.ipynb

Jupyter Project_analysis Last Checkpoint: 2 days ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

```
[1]: import pandas as pd;
import numpy as np;
import matplotlib.pyplot as plt;
import seaborn as sns;
```

```
[2]: df = pd.read_csv("employee_burnout_analysis_dataset.csv");
```

```
[4]: df.head(11)
```

	employee_id	employee_name	department	working_hours_per_day	sleep_hours	stress_level	meetings_per_day	work_mode	productivity_score	salary	leaves_taken
0	EMP1584	Albert Lopez	Marketing	5.2	5.6	10	4	Onsite	56	249127.0	
1	EMP2215	Nicole Young	Sales	12.0	4.8	8	1	Hybrid	73	116462.0	
2	EMP1877	Angela Edwards	Sales	9.3	4.9	6	2	Onsite	92	140851.0	
3	EMP5230	Heidi Padilla	it	8.5	5.4	9	2	Remote	46	226467.0	
4	EMP5159	Charles Porter	HR	8.8	4.1	8	6	Hybrid	74	134054.0	
5	EMP5546	Gabrielle Gilbert	IT	7.1	6.4	9	1	Remote	81	147513.0	
6	EMP3787	George Munoz	Support	12.1	7.8	7	4	Onsite	70	170752.0	
7	EMP2744	NaN	Operations	12.1	8.0	4	5	Remote	93	105172.0	
8	EMP5481	Joseph Bailey	Finance	5.1	7.5	8	6	Remote	65	90133.0	
9	EMP4801	Casey Wilson	Marketing	11.9	8.7	3	1	Remote	55	44702.0	
10	EMP5666	Robert Scott DDS	IT	4.1	4.7	10	4	Remote	71	224180.0	

```
[8]: df.shape
```

2. Data Cleaning Overview

localhost:8888/notebooks/Documents/Employee%20Burnout%20Analysis%20Project/Project_analysis.ipynb

Jupyter Project_analysis Last Checkpoint: 2 days ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

Data Cleaning

```
[10]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5180 entries, 0 to 5179
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   employee_id            5180 non-null   str
1   employee_name          4940 non-null   str
2   department             5054 non-null   str
3   working_hours_per_day  5057 non-null   str
4   sleep_hours            5057 non-null   float64
5   stress_level           5180 non-null   str
6   meetings_per_day       5180 non-null   int64
7   work_mode              5180 non-null   str
8   productivity_score      5180 non-null   str
9   salary                 5056 non-null   str
10  leaves_taken           5180 non-null   int64
11  burnout_risk           5180 non-null   str
dtypes: float64(1), int64(2), str(9)
memory usage: 485.8 KB
```

```
[11]: df.describe()
```

	sleep_hours	meetings_per_day	leaves_taken
count	5057.000000	5180.000000	5180.000000
mean	5.991418	5.97722	15.183977
std	1.732579	3.75385	8.950937
min	3.000000	0.00000	0.000000
25%	4.500000	3.00000	7.000000
50%	6.000000	6.00000	15.000000
75%	7.500000	9.00000	23.000000

3. Fix Data Types and Check Null Values

Home

Project_analysis

localhost:8888/notebooks/Documents/Employee%20Burnout%20Analysis%20Project/Project_analysis.ipynb

Project_analysis Last Checkpoint: 2 days ago

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

Fix DataTypes

```
[12]: df["salary"] = pd.to_numeric(df["salary"], errors="coerce")
[13]: df["working_hours_per_day"] = pd.to_numeric(df["working_hours_per_day"], errors="coerce")
[14]: df["stress_level"] = pd.to_numeric(df["stress_level"], errors="coerce")
[15]: df["productivity_score"] = pd.to_numeric(df["productivity_score"], errors="coerce")
```

Check Null Values

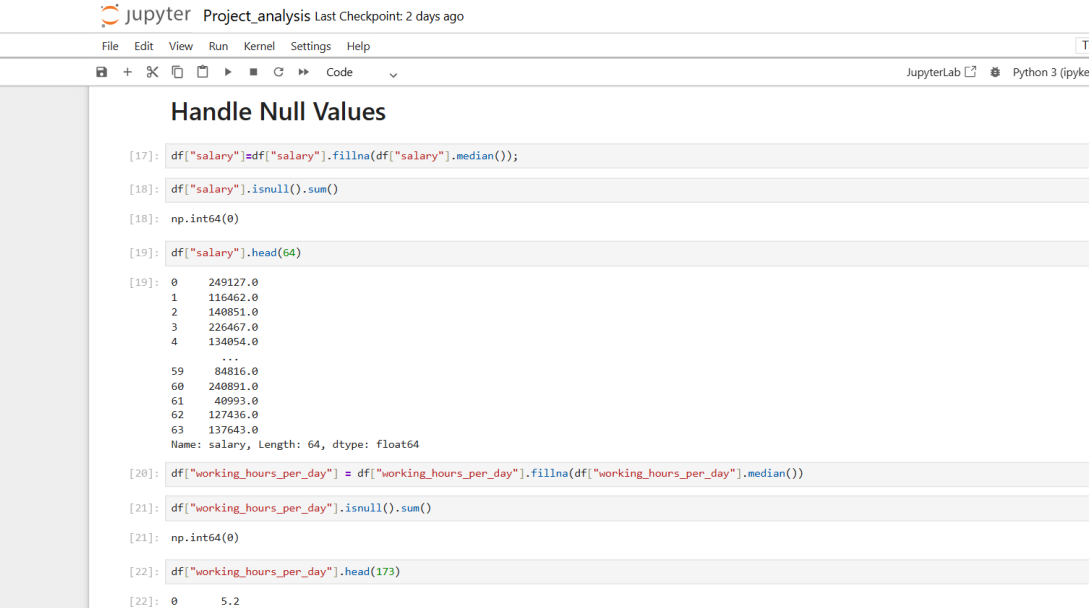
```
[16]: df.isnull().sum()
```

employee_id	0
employee_name	240
department	126
working_hours_per_day	148
sleep_hours	123
stress_level	25
meetings_per_day	0
work_mode	0
productivity_score	25
salary	149
leaves_taken	0
burnout_risk	0
dtype:	int64

Handle Null Values

```
[17]: df["salary"] = df["salary"].fillna(df["salary"].median());
[18]: df["salary"].isnull().sum()
```

4. Handle Missing Values



The screenshot displays a JupyterLab environment with a notebook titled "Project_analysis". The notebook's content area shows a series of code cells and their outputs. The code is focused on data cleaning, specifically handling null values in a DataFrame named 'df'.

The code cells and their outputs are as follows:

- Cell [17]: `df["salary"] = df["salary"].fillna(df["salary"].median());`
- Cell [18]: `df["salary"].isnull().sum()`
- Cell [18]: `np.int64(0)`
- Cell [19]: `df["salary"].head(64)`
- Cell [19]: Output showing the first 64 rows of the 'salary' column. The values range from 0 to 134054.0, with a gap between index 4 and 59. The output is a Series with dtype float64.
- Cell [20]: `df["working_hours_per_day"] = df["working_hours_per_day"].fillna(df["working_hours_per_day"].median());`
- Cell [21]: `df["working_hours_per_day"].isnull().sum()`
- Cell [21]: `np.int64(0)`
- Cell [22]: `df["working_hours_per_day"].head(173)`
- Cell [22]: Output showing the first 173 rows of the 'working_hours_per_day' column. The values range from 0 to 7.6. The output is a Series with dtype float64.

5. Remove Duplicates and Standardize Text

JupyterLab interface showing the execution of code to remove duplicates and standardize text in a dataset.

Remove Duplicate Values

```
[46]: df.duplicated().sum()
[46]: np.int64(159)
[47]: df=df.drop_duplicates()
[48]: df.shape
[48]: (5021, 12)
```

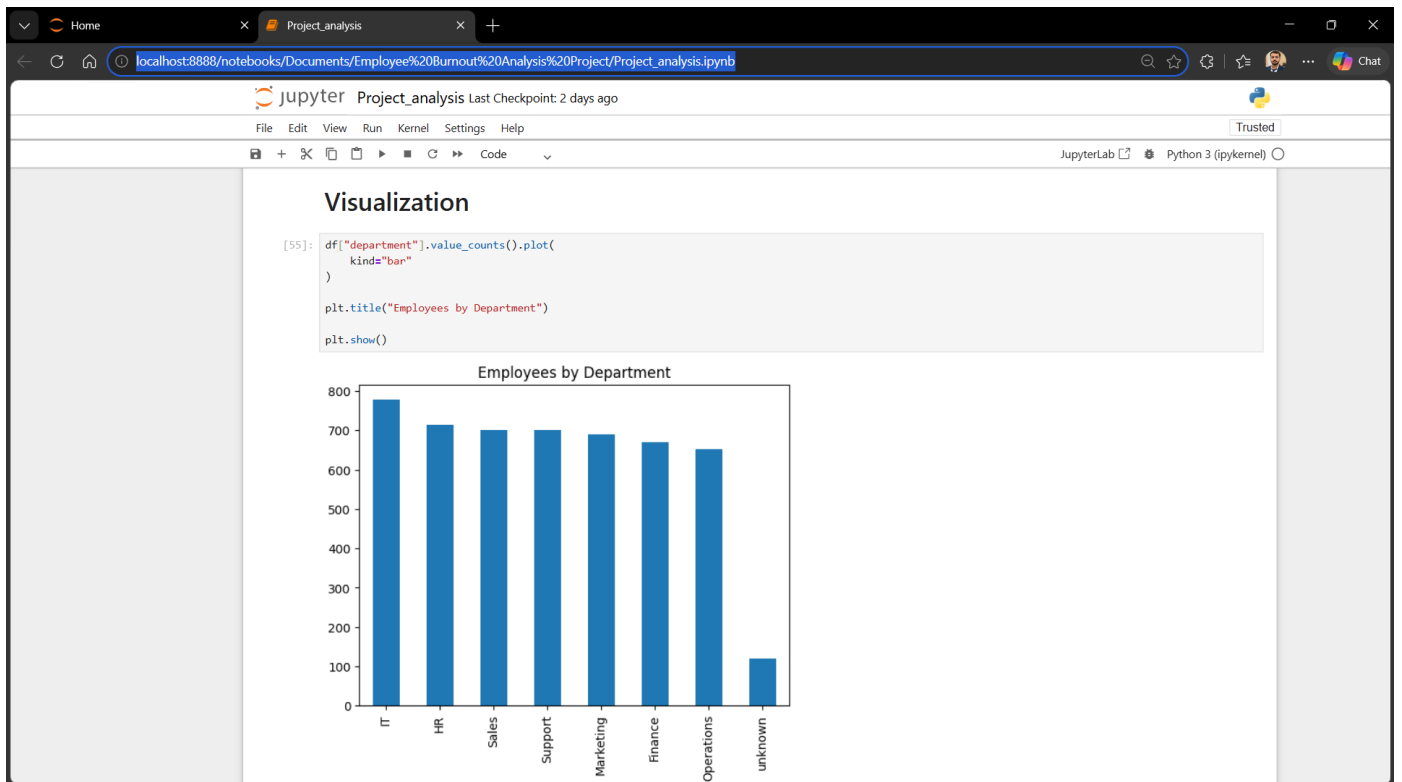
Fix Standardize Text

```
[49]: df["work_mode"] = df["work_mode"].replace({"remote": "Remote", "REMOTE": "Remote"})
[50]: df["department"] = df["department"].replace({"it": "IT"})
```

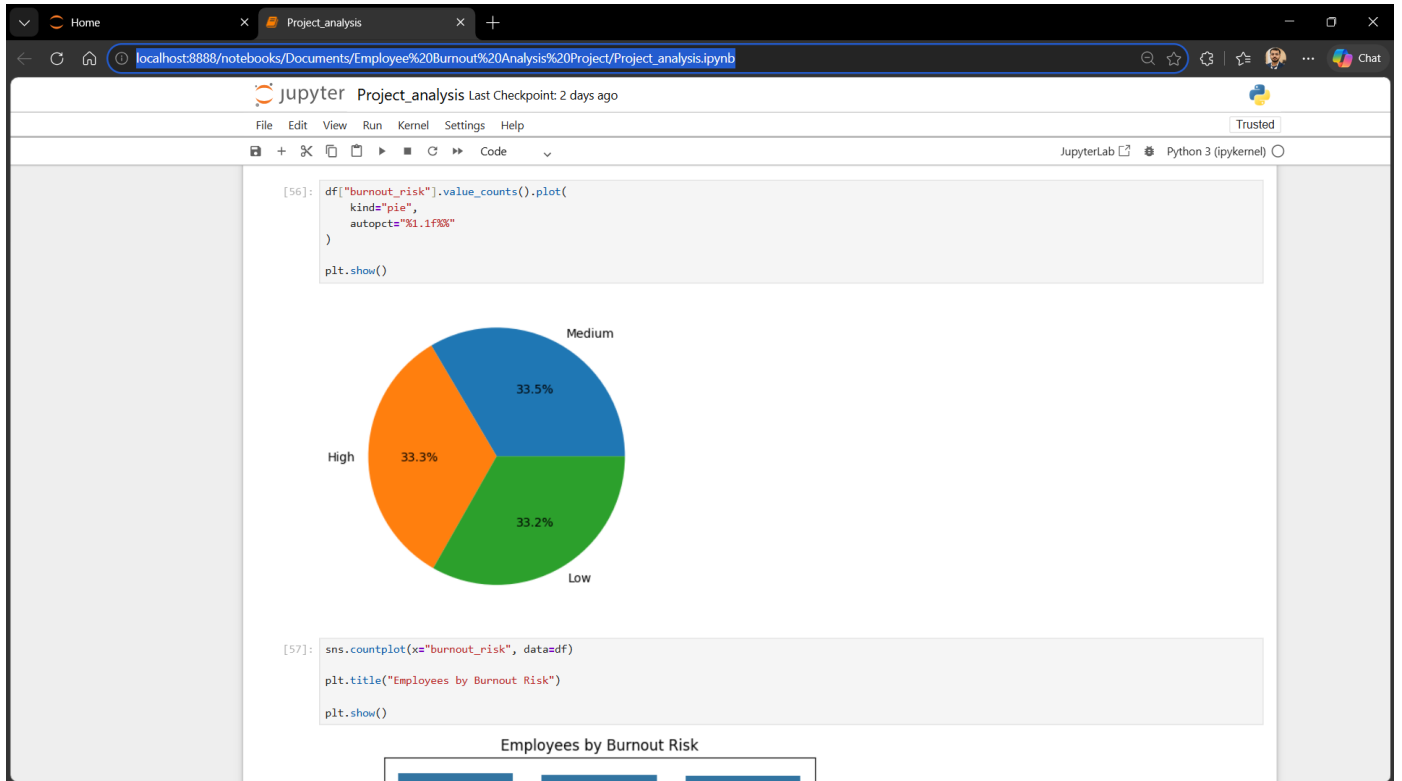
Perform EDA (Exploratory Data Analysis)

```
[51]: df["department"].value_counts()
[51]: department
IT      777
HR      713
Sales   701
Support 701
Marketing 689
Finance 669
Operations 651
unknown 120
Name: count, dtype: int64
[52]: salary_percentage = (df.groupby("department")["salary"].sum() / df["salary"].sum())*100
```

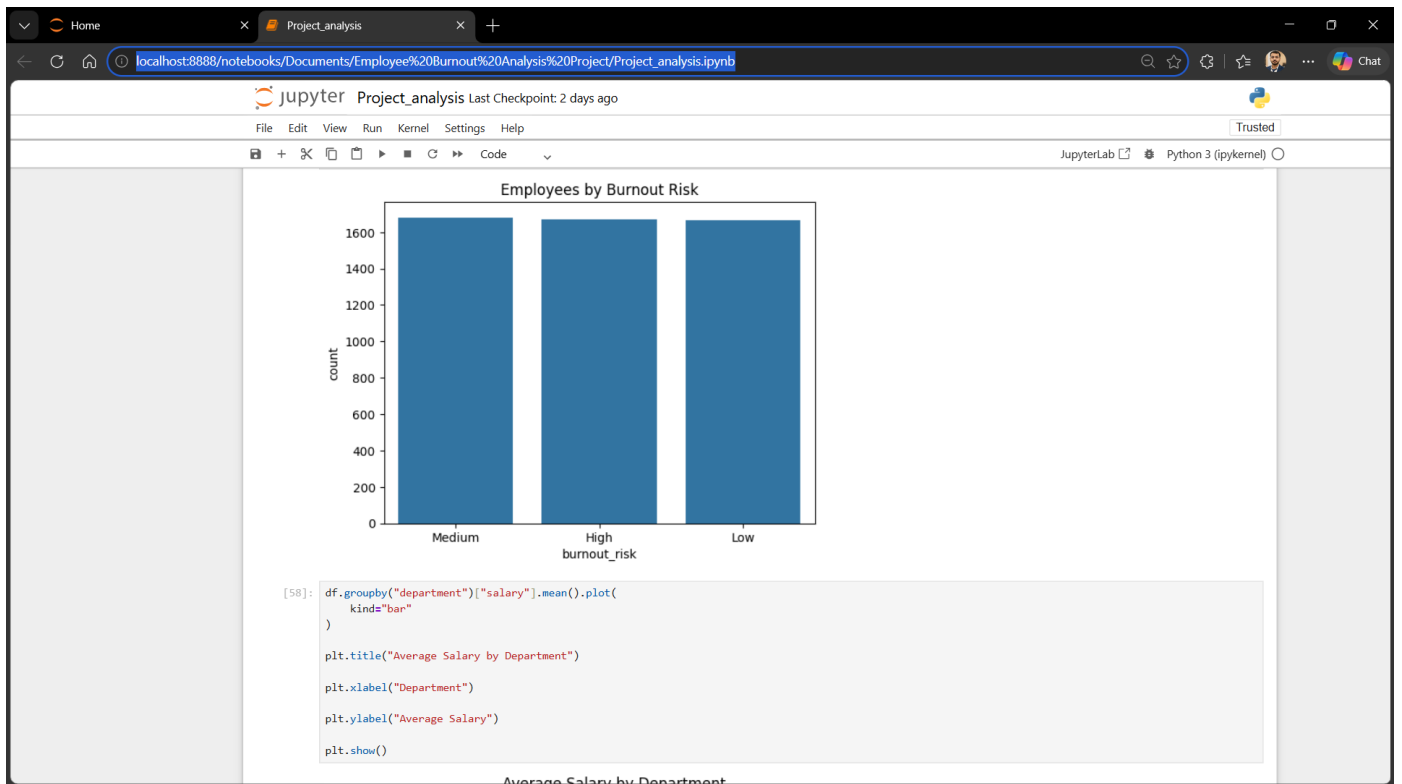
6. Department-wise Employee Distribution



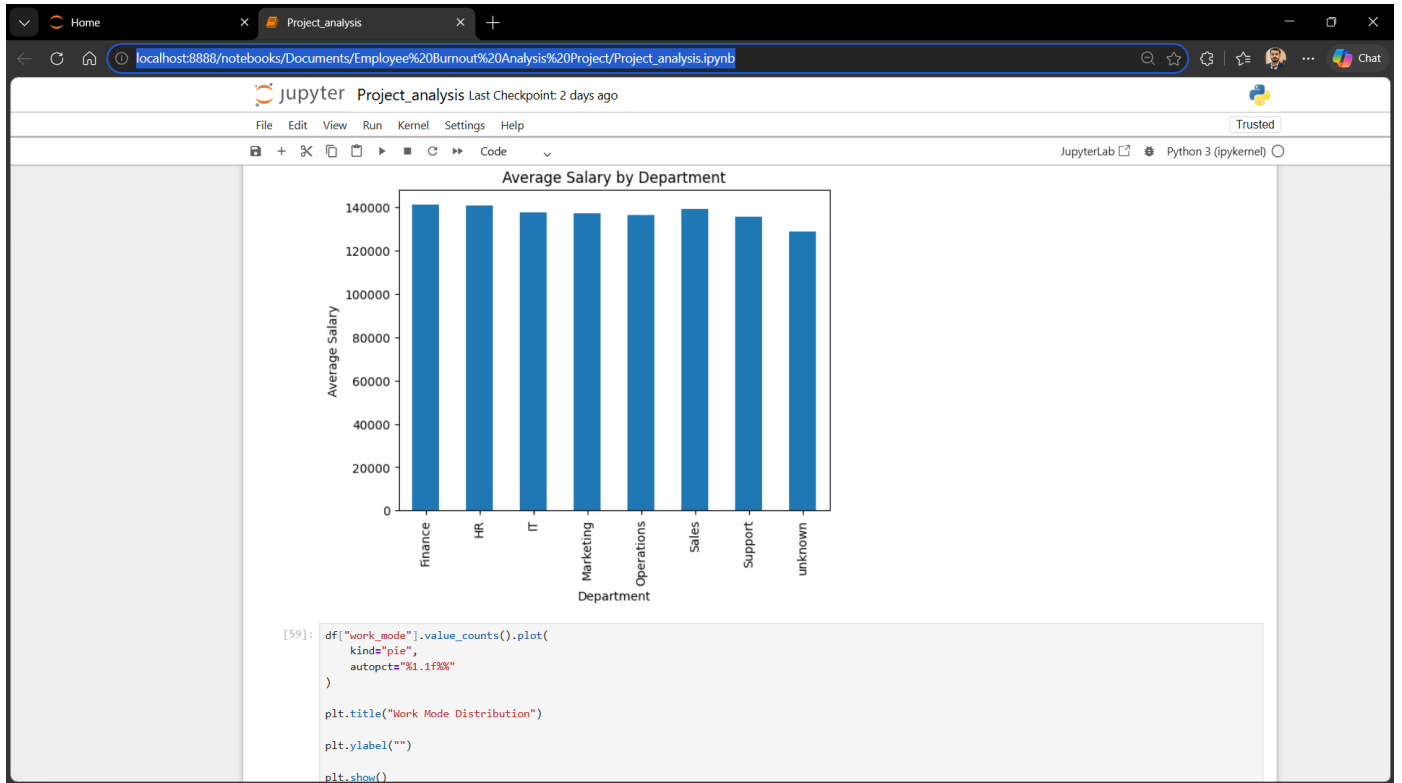
7. Burnout Risk Distribution



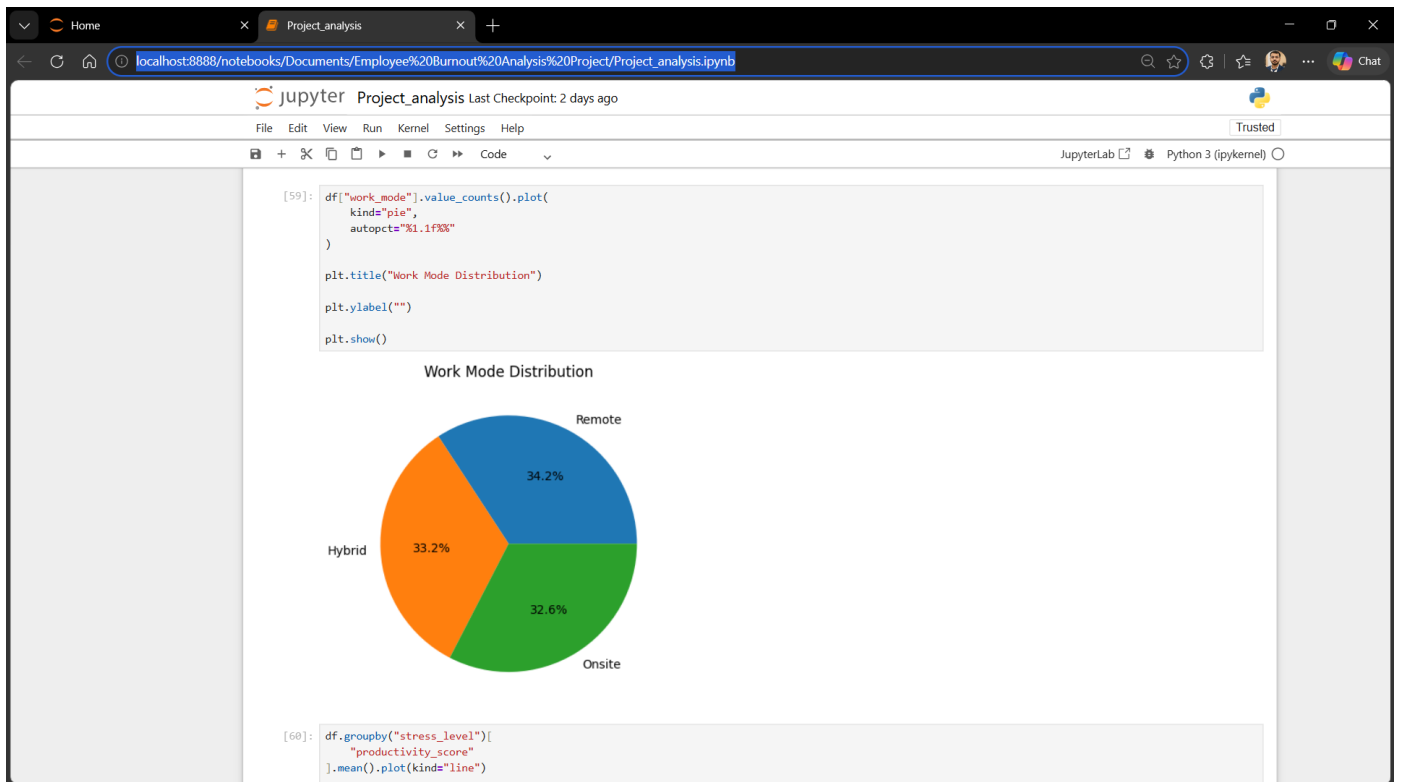
8. Employees by Burnout Risk



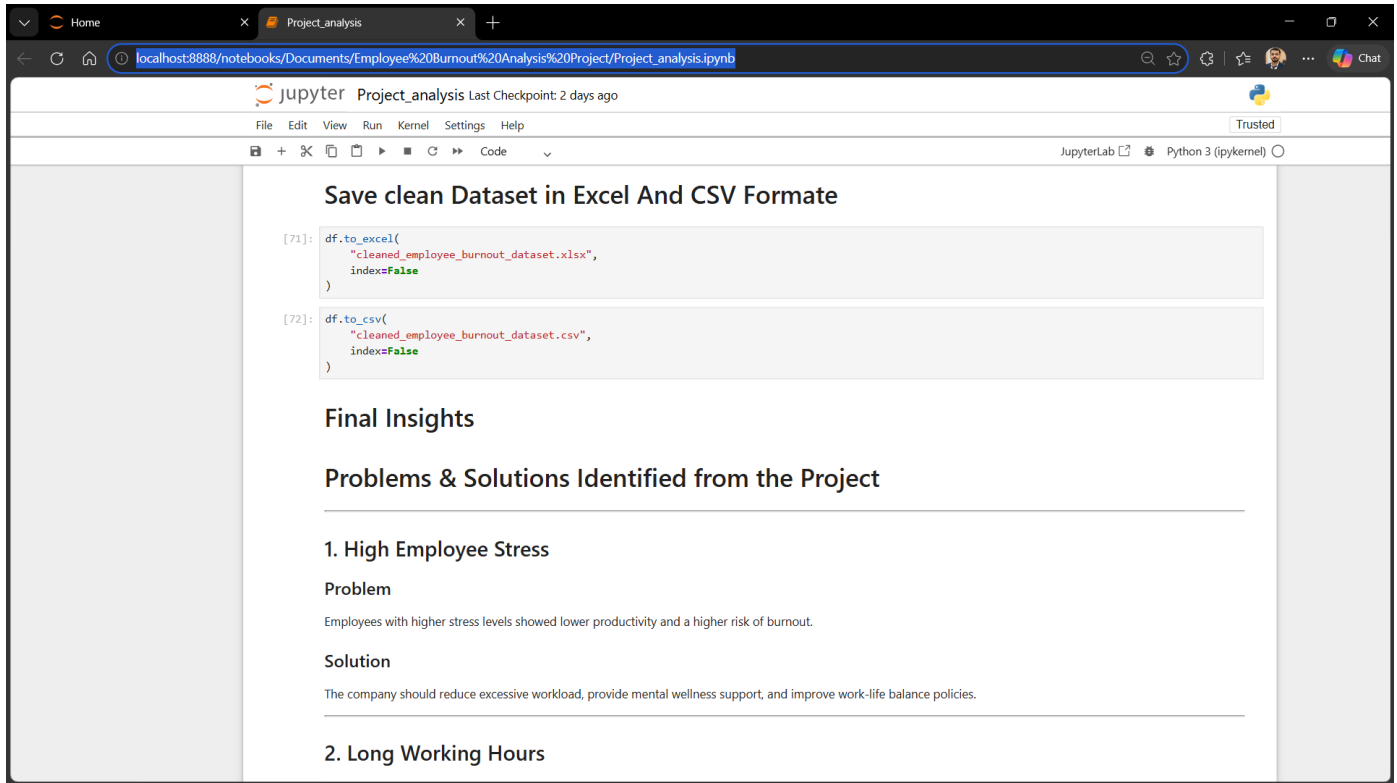
9. Average Salary by Department



10. Work Mode Distribution



11. Save Clean Dataset



The screenshot shows a JupyterLab interface with a notebook titled "Project_analysis". The notebook contains two code cells and a text section. The first code cell uses `df.to_excel()` to save the dataset as `cleaned_employee_burnout_dataset.xlsx`. The second code cell uses `df.to_csv()` to save it as `cleaned_employee_burnout_dataset.csv`. Below the code, the text section is titled "Final Insights" and "Problems & Solutions Identified from the Project". It lists two main issues: "1. High Employee Stress" and "2. Long Working Hours". For each issue, it provides a "Problem" description and a "Solution" recommendation.

```
[71]: df.to_excel(
      1     "cleaned_employee_burnout_dataset.xlsx",
      2     index=False
      3 )

[72]: df.to_csv(
      1     "cleaned_employee_burnout_dataset.csv",
      2     index=False
      3 )
```

Final Insights

Problems & Solutions Identified from the Project

1. High Employee Stress

Problem

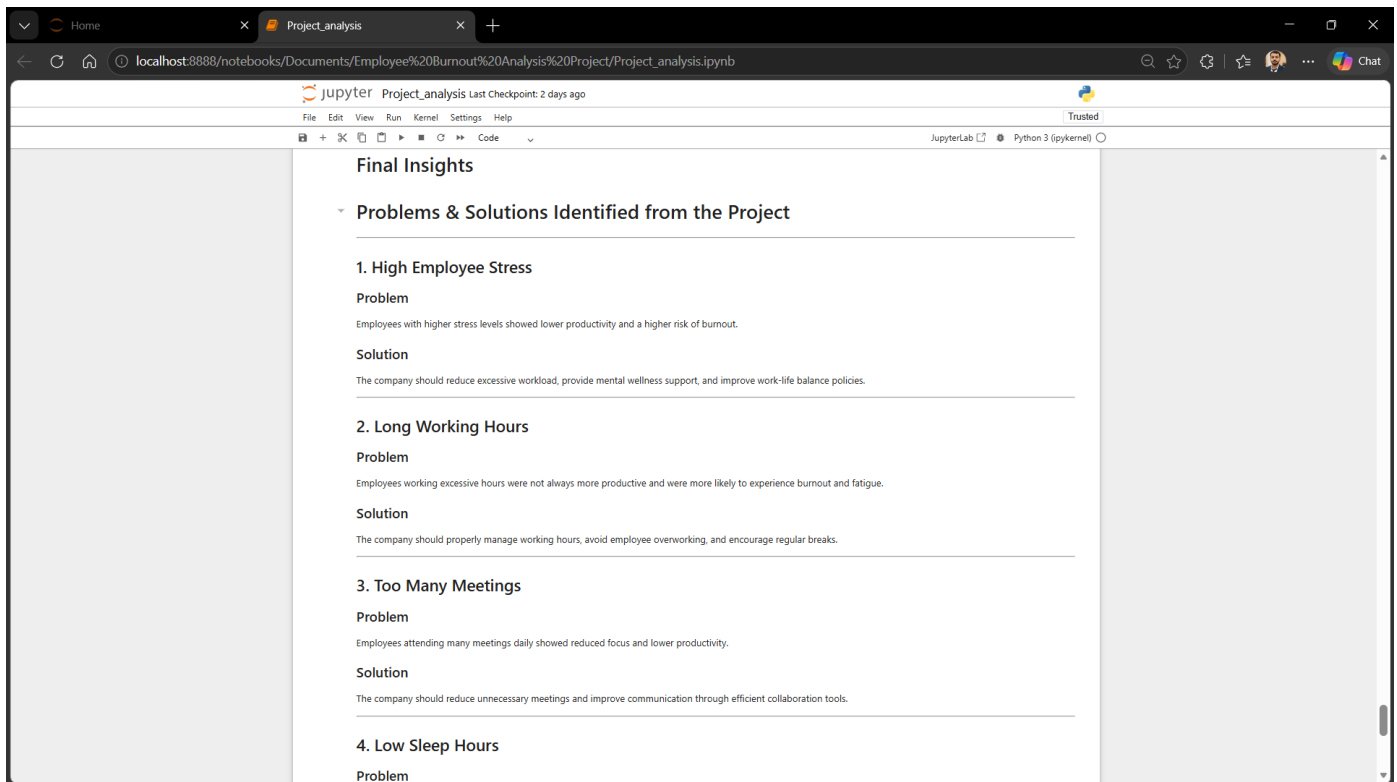
Employees with higher stress levels showed lower productivity and a higher risk of burnout.

Solution

The company should reduce excessive workload, provide mental wellness support, and improve work-life balance policies.

2. Long Working Hours

12. Final Insights and Recommendations



The screenshot shows a JupyterLab interface with a notebook titled "Project_analysis". The notebook contains a text section titled "Final Insights" and "Problems & Solutions Identified from the Project". It lists four main issues: "1. High Employee Stress", "2. Long Working Hours", "3. Too Many Meetings", and "4. Low Sleep Hours". For each issue, it provides a "Problem" description and a "Solution" recommendation.

Final Insights

Problems & Solutions Identified from the Project

1. High Employee Stress

Problem

Employees with higher stress levels showed lower productivity and a higher risk of burnout.

Solution

The company should reduce excessive workload, provide mental wellness support, and improve work-life balance policies.

2. Long Working Hours

Problem

Employees working excessive hours were not always more productive and were more likely to experience burnout and fatigue.

Solution

The company should properly manage working hours, avoid employee overworking, and encourage regular breaks.

3. Too Many Meetings

Problem

Employees attending many meetings daily showed reduced focus and lower productivity.

Solution

The company should reduce unnecessary meetings and improve communication through efficient collaboration tools.

4. Low Sleep Hours

Problem